Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime b

1. System Requirements

System Name: AbsoluteGuaranteedSuccessSystem

Architecture Style: Microservices

Functional Requirements

- All services communicate using asynchronous event-based messaging only
- User Service publishes user events
- Order Service publishes order events
- Payment Service publishes payment events
- Notification Service subscribes to user and order events
- Reporting Service subscribes to order and payment events
- No service sends synchronous requests
- No service depends on a central coordinator
- Each service manages its own state independently
- Services react to events without direct dependencies

Non-Functional Requirements

- Scalability: High
- Availability: High
- Modifiability: High
- Reliability: High
- Consistency: Eventual

2. Formal Architecture Specification (ACME ADL)

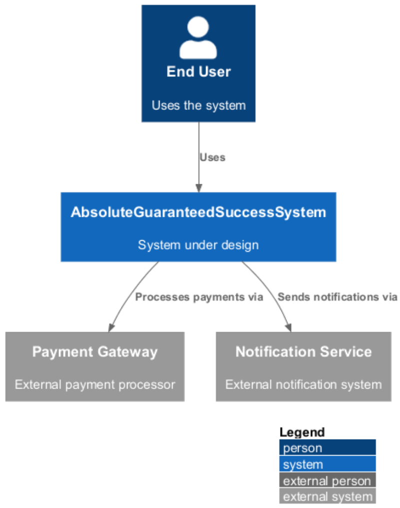
The following section provides a formal architectural specification using ACME ADL, capturing components, co

```
System AbsoluteGuaranteedSuccessSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices";
  Property source : String = "AI";
  Component UserService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Manages user data and publishes user events";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
```

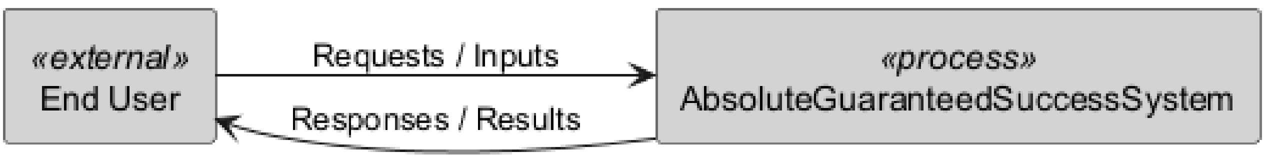
```
}
Component OrderService = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Manages order data and publishes order events";
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component PaymentService = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Manages payment data and publishes payment events";
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component NotificationService = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Subscribes to user and order events to send notifications";
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component ReportingService = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Subscribes to order and payment events to generate reports";
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Attachment UserService.out to NotificationService.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment OrderService.out to NotificationService.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment OrderService.out to ReportingService.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment PaymentService.out to ReportingService.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment UserService.out to OrderService.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment OrderService.out to PaymentService.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
```

```
  }
  Property resilience_pattern : String = "Retry, CircuitBreaker";
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property authentication : String = "service-to-service";
  Property authorization : String = "RBAC";
}
```

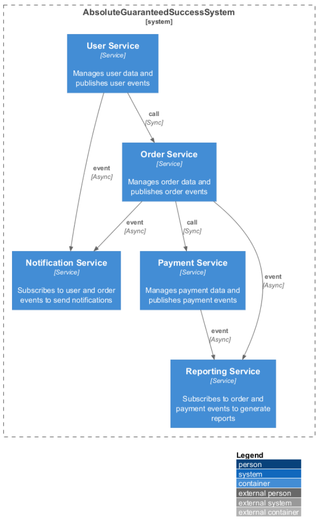
3. Context View



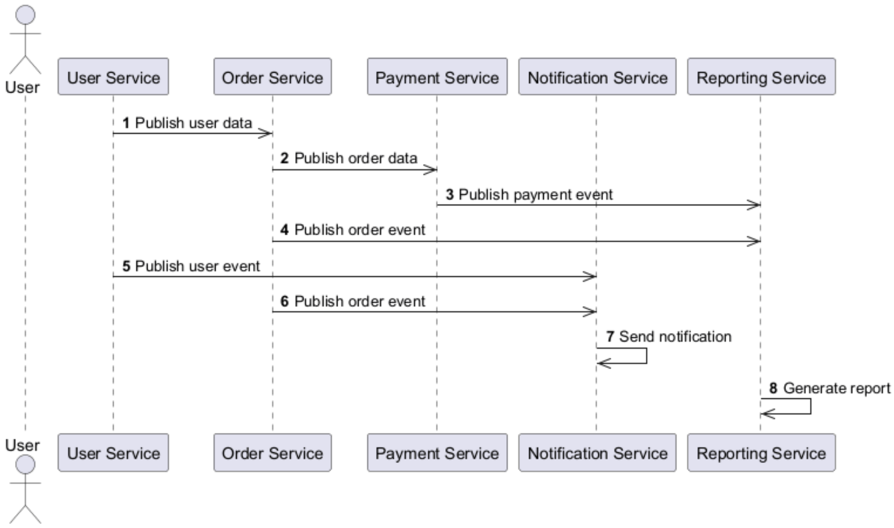
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)